1 Interface overview

HTTPS, mandatory mutual TLS, Arrowhead identity-token header and JSON request/response documents. General management is sysop/whitelist protected; enrollment endpoints use approved certificate fingerprints and profile/serial checks instead of ordinary identity tokens. Error responses contain errorMessage, errorCode, origin and timestamp. Exact DTO and status/error parity is not certified.

1.1 HTTPS interface operations

The listener is HTTPS with JSON by default; raw configuration and CRL responses use their declared media types. A request body is capped at 1 MiB. The table records implemented routes, not a claim of complete behavioral conformance.

Method	Path	Access
GET	/plantdescriptionengine/mgmt/sync-status	management
GET	/plantdescriptionengine/mgmt/plant-descriptions	management
GET	/plantdescriptionengine/mgmt/plant-descriptions/{id}	management
POST	/plantdescriptionengine/mgmt/plant-descriptions	management
PUT	/plantdescriptionengine/mgmt/plant-descriptions/{id}	management
PATCH	/plantdescriptionengine/mgmt/plant-descriptions/{id}	management
DELETE	/plantdescriptionengine/mgmt/plant-descriptions/{id}	management
GET	/plantdescriptionengine/monitor/plant-descriptions	identity
GET	/plantdescriptionengine/monitor/plant-descriptions/{id}	identity
GET	/plantdescriptionengine/monitor/alarms	identity
GET	/plantdescriptionengine/monitor/alarms/{id}	identity
PATCH	/plantdescriptionengine/monitor/alarms/{id}	identity
GET	/plantdescriptionengine/monitor/echo	certificate
GET	/plantdescriptionengine/general/mgmt/get-config	management
POST	/plantdescriptionengine/general/mgmt/logs	management

1.2 Important delimitations

This implementation is a development profile, not a certified interchangeable Java replacement. All systems use private JSON snapshots rather than the Java databases. MQTT application APIs, exhaustive DTO normalization and error parity, automatic data migration and production qualification remain outstanding.

- Topology/include composition, named and metadata matching, periodic alarms, and atomic PDE-to-DSO snapshot synchronization are implemented. Missing producers retain managed empty scopes; nonnegative priorities rank eligible providers. Java SQL import, complete DTO normalization and production qualification remain outside this extension.